DHS/MIS malaria prevalence and child mortality analysis

Main results

- **Higher malaria prevalence is associated with higher post-neonatal mortality, and this holds regardless of covariate adjustment.** Each 10 percentage-point increase in PfPR2-10 is associated with about a 7.5% increase in post-neonatal mortality (ridge-linear summary; 95% CI 4.2%–10.9%; `p<0.001`). The estimate is essentially unchanged whether we use the regional covariate block alone or the full regional-plus-national block that now includes national vaccine coverage and child HIV prevalence.
- **Neonatal mortality, the negative control, shows no such association.** The ridge-linear neonatal estimate is +1.3% per 10 PfPR points (95% CI −0.7% to +3.4%; `p=0.20`), and the selected prevalence term for neonatal mortality is effectively flat. This, together with the stability under HIV and vaccine adjustment, reduces the concern that the main result simply reflects broad differences in child survival between higher- and lower-prevalence settings.
- **A prevalence-by-time interaction now fits best — a change from the earlier, smaller panel.** The linear PfPR term with a time interaction is selected by AIC ahead of the full `te(PfPR, year)` surface (ΔAIC 3.5), a spline time-interaction (ΔAIC 4.0), the additive spline `s(PfPR)+s(year)` (ΔAIC 6.1) and the linear no-interaction form (ΔAIC 12.9). The interaction is +0.0042 per year (`p=2.4e-05`), where on the 936-region-year panel it had been borderline and the time-stable spline was selected. Population-average malaria-attributable fractions of post-neonatal mortality are 7%, 22% and 34% at 10%, 30% and 50% PfPR2-10 (versus a 1% counterfactual), down from 14%, 34% and 40%. Because the historical trend depends strongly on whether a time interaction is admitted, this selection matters more than the modest change in the dose-response itself.
- **The preferred model estimates about 543,000 malaria-attributable post-neonatal deaths in 2024 — the same order as IHME and WHO.** The 95% uncertainty interval is 336,000–716,000, propagating both the fitted attributable-fraction (model-parameter) uncertainty and the all-cause post-neonatal mortality uncertainty (from the IHME/GBD relative interval); the 2000 estimate is 640,000. Across the three time structures the 2024 total ranges from 477,000 to 547,000. The point estimate is about 27% higher than the IHME/GBD 2024 estimate (about 428,000) and about 22% higher than an approximate WHO under-five figure (about 446,000); the intervals overlap. This is not a validation against a common estimand: the model captures direct and indirect malaria-associated post-neonatal mortality, whereas IHME and WHO assign deaths specifically to malaria.

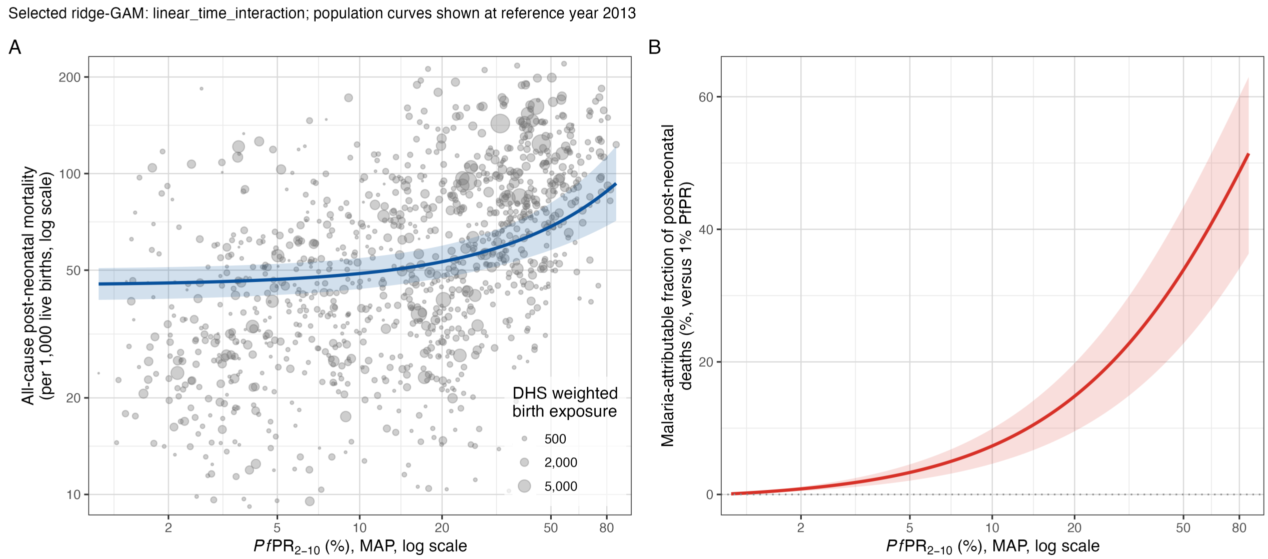

Figure 1: Malaria and post-neonatal mortality. **A:** all-cause post-neonatal mortality versus PfPR2-10, with point size proportional to the survey region's weighted birth exposure and the fitted ridge-GAM curve. **B:** the model-estimated malaria-attributable fraction of post-neonatal deaths as a function of PfPR2-10.

The sections below give the data, methods and supporting detail.

Data and methods

Sample. The unit of analysis is a DHS/MIS survey-region-year (Births Recodes, sub-Saharan Africa, 2000–2024; AIS excluded). Each region receives the population-weighted MAP PfPR2-10 estimate for the survey year; regional all-under-five and neonatal mortality come from birth histories via the DHS synthetic-cohort life-table method, and post-neonatal mortality is their difference. The assembled panel has 1,125 region-years (119 surveys, 36 countries); the primary sample requires country-mean PfPR2-10 above 1% and regional PfPR2-10 at least 1%, plus positive outcomes/exposure, giving **1,108 region-years (117 surveys, 35 countries)**. The full data flow is shown in the analysis-flow diagram.

Panel coverage. The panel is now assembled from the DHS survey registry rather than inherited from an earlier aggregate. Three faults in that path were repaired — a Windows-1250 mis-decoding of accented region names, an empty `v022` that made `chmort` abort for four surveys, and a MAP regional extraction that had never completed — and recode region names are now reconciled against boundary names across language, word order, spelling, qualifier clauses and granularity. Together these took the panel from 936 region-years (106 surveys) to 1,125 (119 surveys).

This matters for interpretation, not only for coverage: the linear no-interaction estimate fell at every step as coverage improved (+8.5%, +8.30%, +7.61%, +7.33% per 10 PfPR points across the four successive panels), which suggests the surveys previously dropped were not missing at random with respect to the association. Six regions across two surveys still cannot be joined — Mali 2012 did not survey four of its regions, and Uganda 2016's Buganda/Central labelling is genuinely ambiguous — and per-survey merge detail is recorded in `results/dhs_rebuild/region_merge_quality.csv`.

Covariates. Twenty-five candidate covariates are screened before imputation; those with at most 5% missingness are retained and singly imputed (country median, overall-median fallback). The **18 retained covariates** enter a single standardised, ridge-penalised block (proportions logit-transformed):

Level	Covariates
Survey-region (DHS)	urban residence, DTP3, measles, facility delivery, maternal education, exclusive breastfeeding, short birth interval, maternal age at first birth, improved water, improved sanitation, household electricity
National, exact year	WUENIC Hib3, PCV-completion, rotavirus-completion; and child (0–14) HIV prevalence (log)
National, nearest year	log GDP per capita, log health expenditure per capita, political stability

Seven candidates are excluded for excess missingness (the three direct DHS vaccine completion recodes, household wealth, and the three anthropometry measures). Political stability is now retained: the World Bank had archived the `PV.EST` indicator, so the series was silently empty and the covariate was dropped for 100% missingness; it is read from the current `GOV_WGI_PV_EST` series. National DTP3, electricity access and health-expenditure-share are dropped as redundant with their regional or per-capita counterparts.

Child HIV prevalence is derived from the UNAIDS 2025 estimates workbook (number of children 0–14 living with HIV) divided by the World Bank 0–14 population (`SP.POP.0014.TO`), joined by ISO3 and exact survey year, and entered on the log scale because it spans a very wide range across countries (country means from about 0.02% in Madagascar to about 3.8% in Eswatini). Nigeria and Comoros publish only a 15–19 series (4% of the sample) and are imputed with a provenance flag. Source choices are in `R_dhs/COVARIATE_SOURCES.md`.

Model. Approximate death counts (rate × birth exposure) are modelled with a negative-binomial GAM: `offset(log(exposure))`, the ridge covariate block, country random intercepts and country-specific random linear PfPR slopes. Five PfPR/time specifications are compared by ML/AIC on the same 1,108 observations, and the selected form is refitted by REML for post-neonatal, all-under-five and neonatal mortality.

Association and model selection

From the ridge-linear summary models, each 10 percentage-point increase in PfPR2-10 is associated with:

Outcome	Change in mortality	95% CI	p-value
Post-neonatal	+7.49%	+4.17% to +10.90%	<0.001
All under-five	+5.26%	+2.73% to +7.86%	<0.001
Neonatal (negative control)	+1.34%	−0.72% to +3.45%	0.20

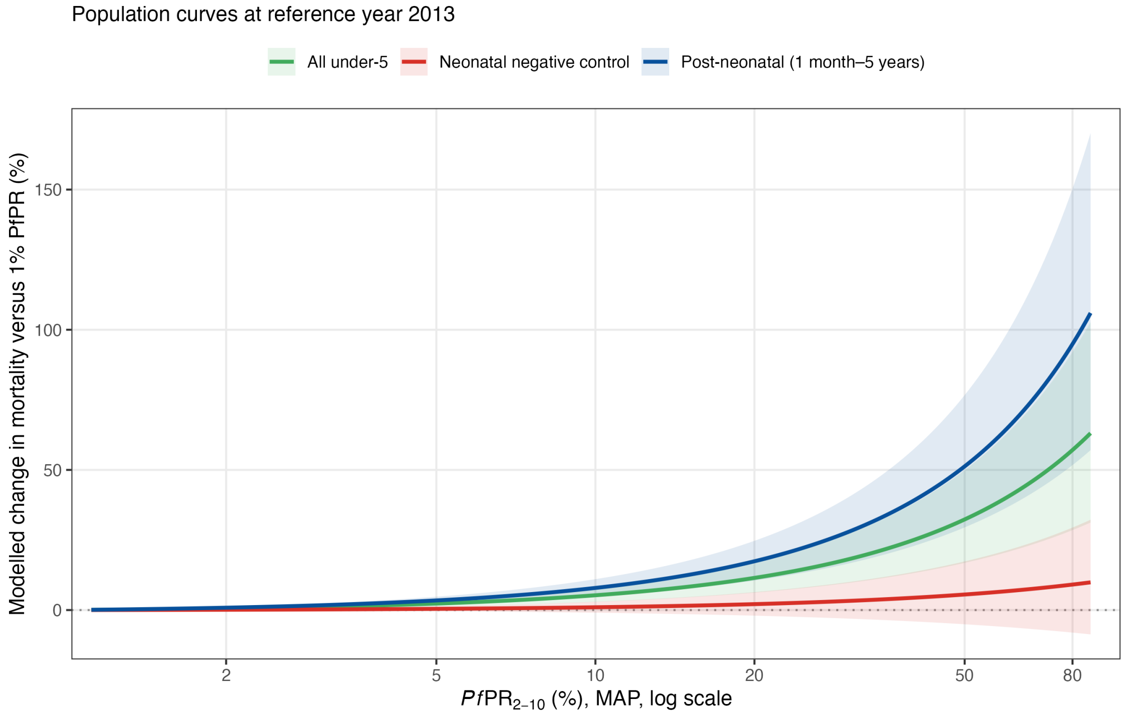

Figure 2: Modelled change in mortality versus 1% PfPR for the three outcomes. The post-neonatal and all-under-five associations rise with prevalence; the neonatal negative control stays flat and its interval spans zero.

Five specifications are compared by AIC on the 1,108-observation post-neonatal sample:

Model	Specification	AIC	ΔAIC
Linear, interaction	`PfPR + PfPR × year + s(year)`	8556.01	0.00
Full tensor surface	`te(PfPR, year)`	8559.47	3.46
Spline, `ti` interaction	`s(PfPR) + s(year) + ti(PfPR, year)`	8560.01	4.00
Spline, no interaction	`s(PfPR) + s(year)`	8562.08	6.07
Linear, no interaction	`PfPR + s(year)`	8568.90	12.89

An independent AIC comparison for neonatal mortality selects an effectively linear, flat prevalence term; it selects the linear no-interaction form, whose prevalence coefficient is null (`p=0.21`), consistent with the negative-control table above.

The selected specification gives the following malaria-attributable fractions of post-neonatal mortality (country random effects set to zero; reference year 2013):

PfPR2-10	Attributable fraction	95% interval
10%	7.3%	4.6%–9.9%
30%	21.7%	14.2%–28.6%
50%	33.9%	22.8%–43.4%

Robustness

Robustness is the central message: the association barely moves across covariate blocks, samples, model structures and likelihood.

- **Covariate adjustment.** Child HIV prevalence is itself the strongest single predictor of post-neonatal mortality in the block (+14.1% per standard deviation), yet adding it leaves the malaria estimate essentially unchanged. The regional block alone and the full regional-plus-national block give a similar attributable fraction (22.5% and 21.5% at 30% PfPR).

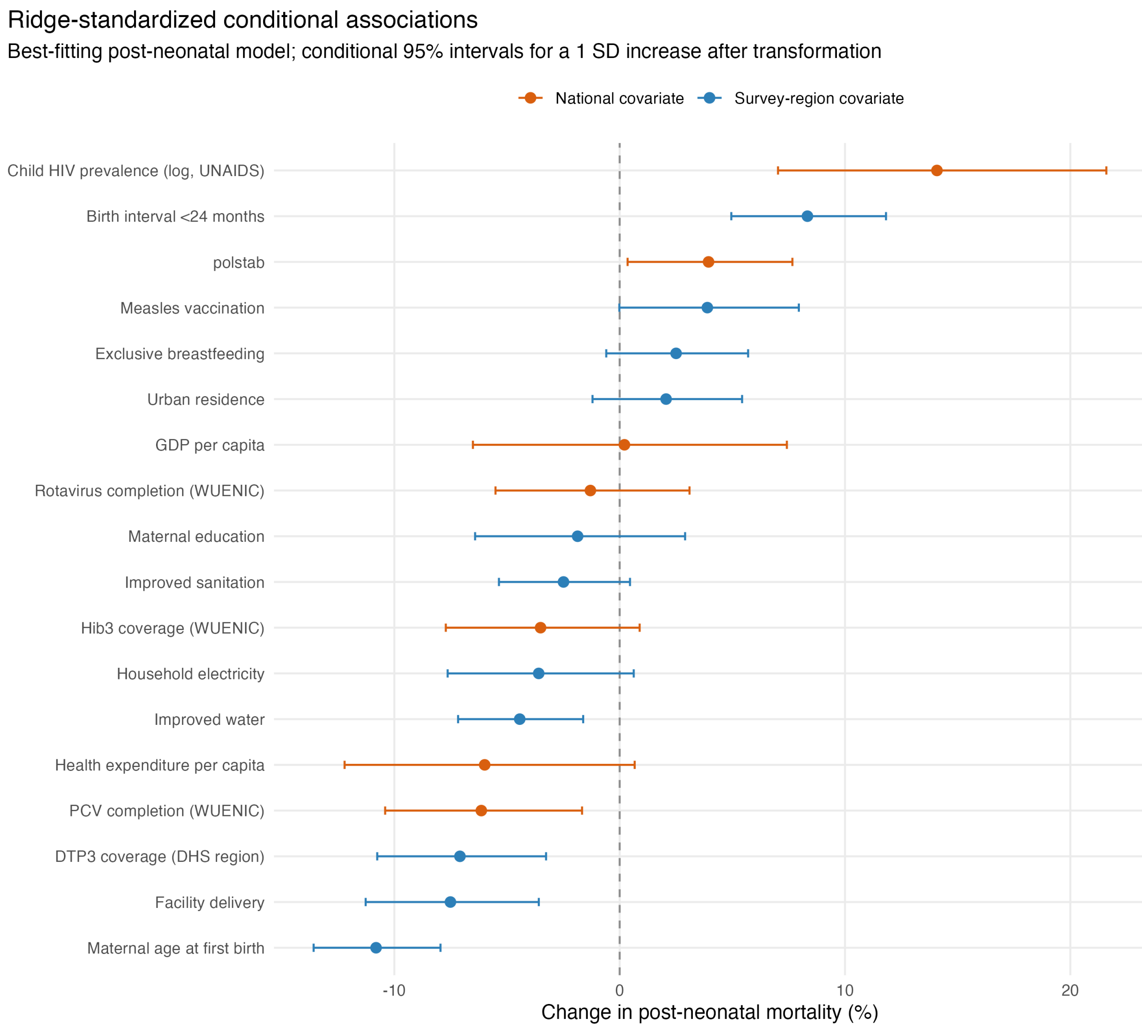

Figure 3: Ridge-standardized conditional associations for the 17 covariates in the best-fitting post-neonatal model (percentage change per 1 SD after transformation; ridge-shrunk, not independent causal effects).

- **Samples, structure and likelihood.** The attributable fraction at 30% PfPR stays near 22% across alternative samples (21.4% including sub-1% regions, 25.8% on the 5–40% range, 23.1% on complete cases) and across model-structure choices (18.7% without country-specific PfPR slopes, 22.5% with regional covariates only); it rises materially only in a national-covariates-only model that fits far worse (33.6%). A log-Gaussian rate model is less precise and its interval spans zero (+5.3% per 10 PfPR points, 95% CI −1.6% to +12.8%; `p=0.14`).

External validation: bednet trials

An independent triangulation against eight insecticide-treated-net/curtain cluster-RCTs — which measured both parasite prevalence and child mortality — gives a slope of the same order as the observational model: a weighted meta-regression gives about an 8.6% mortality reduction per 10-point drop in age-standardised PfPR2-10, against the model's ridge-linear 7.5% (the trial estimate is imprecise — `p=0.12`, eight heterogeneous trials — and it is unchanged by the panel rebuild, being wholly external to it). Predicted and observed trial effects broadly track each other.

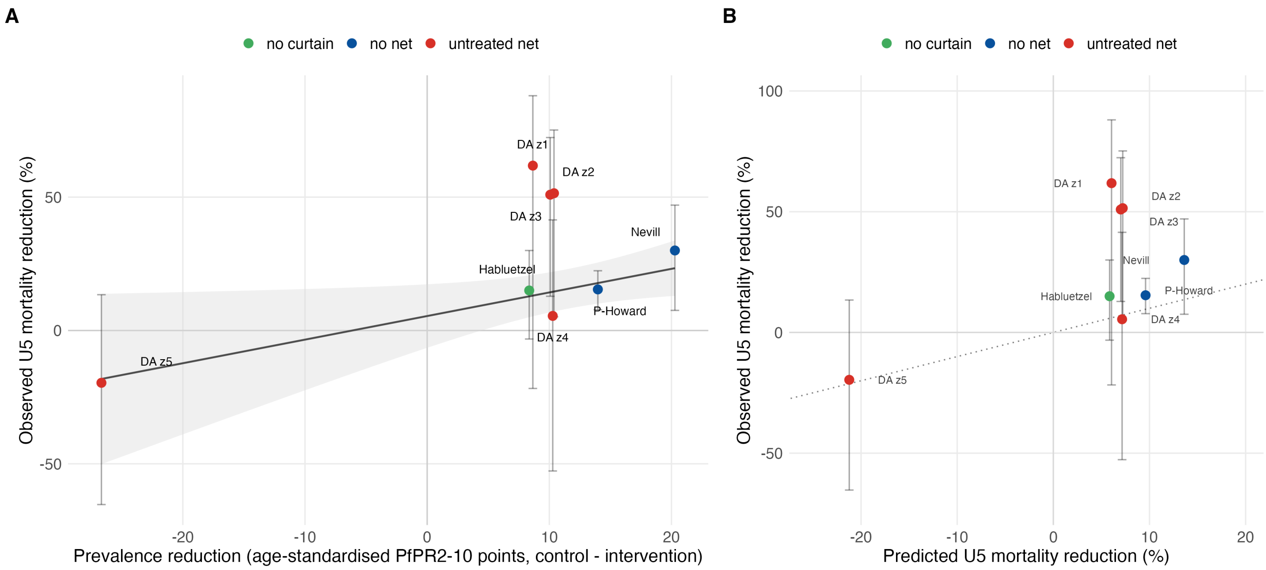

Figure 4: RCT triangulation. **A:** observed under-five mortality reduction versus the age-standardised prevalence reduction across trial arms (weighted fit). **B:** mortality reductions predicted from the model's prevalence slope versus observed (dotted line = 1:1).

National burden

Latest-year comparison

The comparison is not perfectly contemporaneous: the model surface is evaluated at 2024 using the latest available 2024 MAP PfPR, IGME all-cause mortality and live births for 43 matched sub-Saharan African countries; IHME is 2024 under-five malaria deaths; WHO WMR 2025 reports all-age country deaths for 2024, so the under-five comparator applies WHO's ~75% age-share statement.

Source	Deaths
Selected model (full `te` surface)	**543,000**
Spline, `ti` interaction	547,000
Spline, no time interaction	477,000
IHME/GBD SSA under-five, 2024	428,000
WHO African-region all-age, 2024	579,000
WHO African-region under-five proxy, 2024	434,000

Temporal trend

The historical trend is the quantity most sensitive to whether a time interaction is admitted, and on the rebuilt panel the interaction is now favoured. Applied to national PfPR and all-cause mortality series, estimated post-neonatal malaria deaths from 2000 to 2024 fall by:

Model	2000	2024	Change
Full `te` surface (selected)	640,000	543,000	−15%
Spline `ti` interaction	693,000	546,000	−21%
Spline, no time interaction	961,000	476,000	−50%

The spread across these three structures is the headline uncertainty in the trend, and it is wide: a time-stable attributable fraction implies the burden halved over the period, whereas the time-varying structures — which let the attributable fraction rise gradually — imply a decline of only 15–21%. On the 936-region-year panel the time-stable form was selected and the −52% figure was the headline; on the fuller panel it is the worst-fitting of the five specifications, so the smaller declines should now carry more weight.

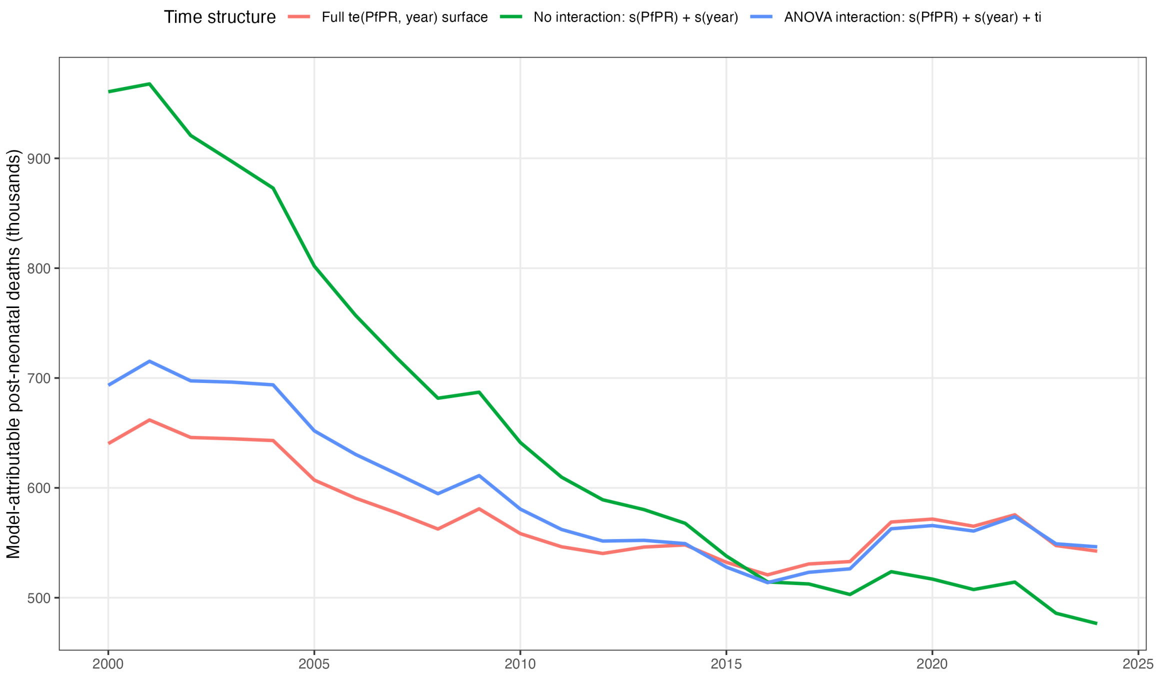

Figure 5: National malaria-attributable post-neonatal deaths over time under the three time structures.

Country differences from IHME

The aggregate is now about 27% above IHME (model 543,000 vs IHME 428,000 in 2024, a ratio of 1.27), and the country-level differences are larger still (AIC-selected full `te` surface; ten countries with the largest absolute difference from IHME/GBD, all for 2024):

Country	Model	IHME 2024	Model/IHME
DR Congo	126,000	61,800	2.05
Nigeria	200,000	138,100	1.45
Chad	13,400	4,900	2.73
Mozambique	14,900	10,300	1.44
Benin	9,900	7,400	1.34
Guinea	9,500	7,100	1.35
Niger	22,700	20,300	1.11
Central African Republic	6,100	3,900	1.58
South Sudan	6,600	4,800	1.38
Togo	2,800	1,900	1.48

The pattern differs from the earlier panel in a way worth noting: previously the model's excess over IHME in DR Congo and Nigeria was offset by lower estimates elsewhere, which kept the aggregates close. On the rebuilt panel the ten largest differences all run in the same direction, so the aggregate sits well above IHME rather than near it. DR Congo alone accounts for about 65,000 of the 115,000 aggregate excess. (IHME/GBD malaria estimates are unavailable for Sudan and South Africa, so the IHME total covers 41 of the 43 countries; both contribute negligibly.)

Interpretation and limitations

The data provide fairly robust evidence of a positive cross-sectional association between malaria prevalence and post-neonatal mortality that survives adjustment for national vaccine rollout and child HIV prevalence, the neonatal negative control, alternative samples, and model-structure choices. The main remaining limitations are:

- the aggregate burden is not a validation against a common estimand — the model includes indirect malaria-associated deaths, whereas IHME and WHO assign deaths specifically to malaria, and national extrapolation uses population-average effects that ignore country random effects;
- 2025 exposure and all-cause mortality inputs are unavailable, and the WHO under-five comparison applies a regional 75% age-share to every country;
- child HIV prevalence is derived from UNAIDS 0–14 counts and a World Bank denominator rather than a published rate, and Nigeria and Comoros (about 4% of the sample, including the largest-burden country) lack an under-15 series and are imputed;
- the log-Gaussian likelihood sensitivity is positive but its interval spans zero, so the exact effect size remains uncertain;
- the historical trend is not well identified. The five specifications are separated by only ~13 AIC points yet imply declines from 15% to 50% over 2000–2024, and the selection flipped from the time-stable spline to a time-interaction form when the panel was completed. The trend should be read as a range, not a point;
- the dose-response estimate fell monotonically as survey coverage was repaired (+8.5% to +7.3% per 10 PfPR points across four successive panels). That pattern is not noise-shaped and suggests the previously dropped surveys differed systematically; it is possible that further coverage work would move it again.

Analysis flow

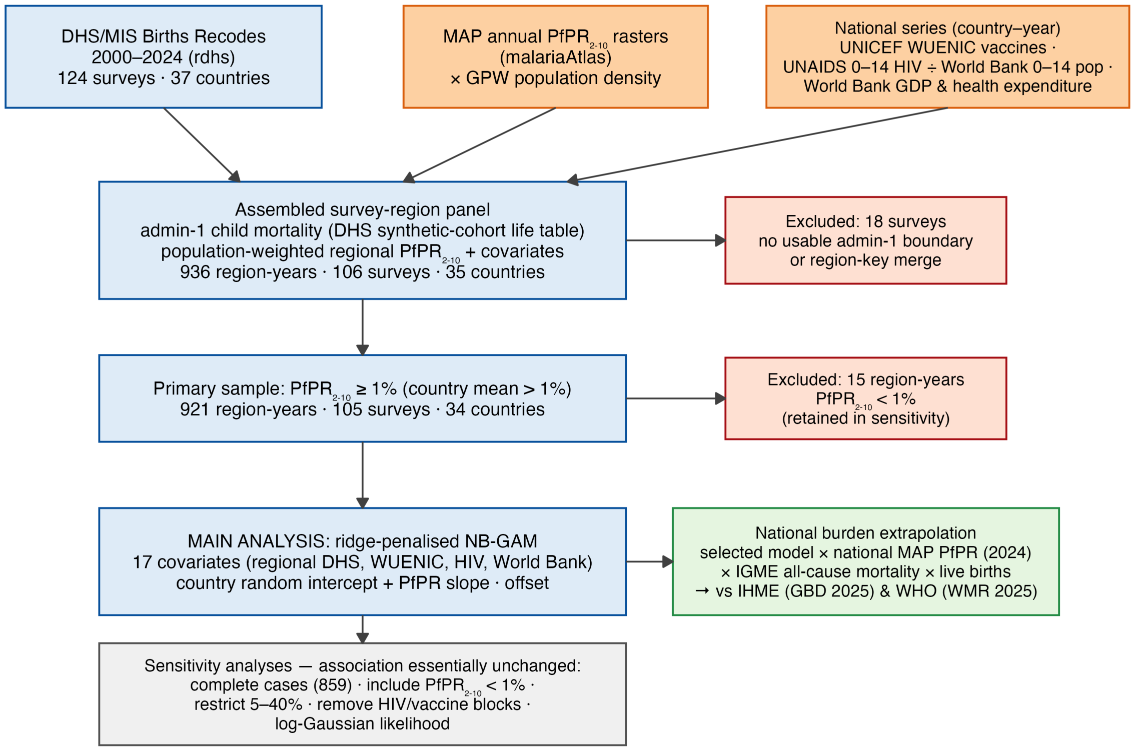

Figure 6: Data and analysis flow. External data (MAP PfPR2-10 rasters, UNICEF WUENIC vaccine coverage, UNAIDS/World Bank child HIV prevalence, and World Bank economic series) feed the assembled survey-region panel, which is filtered to the primary sample, analysed with the ridge-penalised model, and extrapolated to a national burden.